

Feeling of weight in a lift

R = normal reaction = the **reading** shown on the **weighing scale**

W = weight

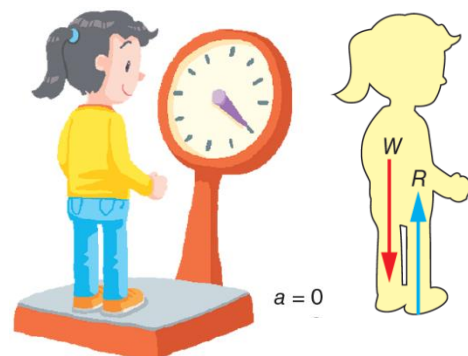
(i) The lift has **no acceleration** under the following 3 situations:

1. the lift is **at rest**
2. the lift is moving upwards with **uniform velocity**
3. the lift is moving downwards with **uniform velocity**

Remarks:

Since normal reaction R = weight mg , there is **no acceleration** for the object and the object has a feeling of **normal weight**.

$$\begin{aligned}\therefore R &= mg \\ \therefore a &= 0\end{aligned}$$



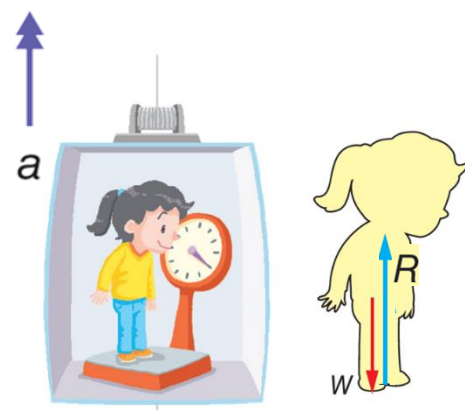
(ii) The lift has **upward acceleration** under the following situations:

1. the lift is moving **upwards** with **acceleration**
2. the lift is moving **downwards** with **deceleration**

Remarks:

Since normal reaction $R > \text{weight } mg$, there is **acceleration** in the upward direction and the object has a feeling of **gain in weight**.

$$\begin{aligned}\text{By } F_{\text{net}} &= ma \\ \therefore R &> mg \\ \therefore R - mg &= ma\end{aligned}$$



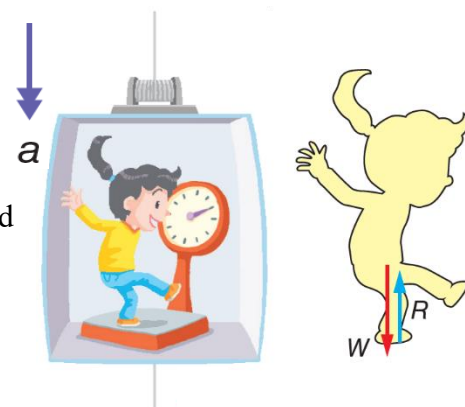
(iii) The lift has **downward acceleration** under the following situations:

1. the lift is moving **downwards** with **acceleration**
2. the lift is moving **upwards** with **deceleration**

Remarks:

Since normal reaction $R < \text{weight } mg$, there is **acceleration** in the downward direction and the object has a feeling of **loss in weight**.

$$\begin{aligned}\text{By } F_{\text{net}} &= ma \\ \therefore mg &> R \\ \therefore mg - R &= ma\end{aligned}$$



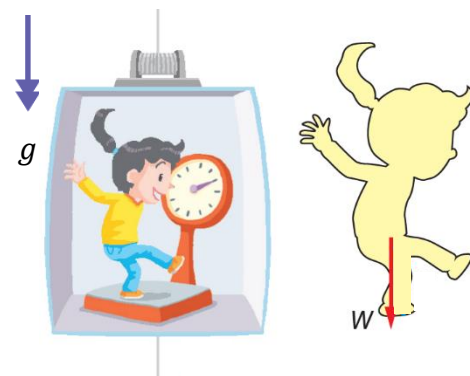
(iv) The lift under **free fall**:

If the rope supporting the lift was broken, the lift would have **free fall under gravity** and hence all object inside the lift also undergo **free fall under gravity**.

Remarks:

The weight of object is **mg** but normal reaction **R** acting on object by the contact surface is **zero**. Hence, the object feels **weightless** and the **balance reading** become **zero** as it is placed under the object.

$$\begin{aligned} \text{By } F_{\text{net}} &= ma \\ \therefore mg &= ma \\ \therefore a &= g \end{aligned}$$



1. A boy of mass 60 kg stands on a weighing scale inside a lift. The scale is graduated in Newtons and reads 588.6 N when the lift is stationary. While the lift travels up from the ground floor to the 10th floor, the scale's reading changes successively to (a) 660 N, (b) 588.6 N, and (c) 570 N. When the lift is stationary on the 10th floor the reading is 588.6 N again. What can you tell about the motion of the lift?

- (a) Accelerating upward
(b) Constant velocity
(c) Decelerating upward

2. Kelvin is standing on a balance inside a lift. The table shows the readings of the balance in three situations.

Motion of the lift	Reading of the balance
moving upwards with a uniform speed	R_1
moving downwards with a uniform speed	R_2
moving upwards with an acceleration	R_3

B

Which of the following relationships is correct ?

- A. $R_1 = R_2 > R_3$ ① When the lift is moving upwards with a uniform speed, there is no acceleration, thus $R_1 = mg$.
 B. $R_3 > R_1 = R_2$ ② When the lift is moving downwards with a uniform speed, there is no acceleration, thus $R_2 = mg$.
 C. $R_1 > R_2 > R_3$ ③ When the lift is moving upwards with an acceleration, there is an upward net force, thus $R_3 > mg$.
 D. $R_3 > R_1 > R_2$ $\therefore R_3 > R_1 = R_2$

3. A man weighs an object with a spring balance in a lift. Before the lift moves the scale reads 50 N. The lift goes down and then stops. The reading on the scale is
- A. more than 50 N when the lift starts, and remains steady until it comes to rest.
 B. less than 50 N when the lift starts, and remains steady until it comes to rest.
 C. more than 50 N as the lift starts, and less than 50 N as it comes to rest.
 D. less than 50 N as the lift starts, and more than 50 N as it comes to rest.

Reading on the balance scale is the normal reaction R .

When the lift starts to go down, it undergoes downward acceleration,

$$\therefore mg - R = ma \quad \therefore R < mg \quad \therefore \text{reading is less than 50 N}$$

When the lift is going to stop, it undergoes downward deceleration (i.e. acceleration upwards)

$$\therefore R - mg = ma \quad \therefore R > mg \quad \therefore \text{reading is more than 50 N}$$

D

4. Diana of mass 50 kg stands on a weighing scale inside a lift. When the lift slows down uniformly at 1 m s^{-2} in the upward journey, what is the magnitude of the force acting on the weighing scale by Diana?

- A 0
B 441 N
C 491 N
D 541 N

Take downward as positive.

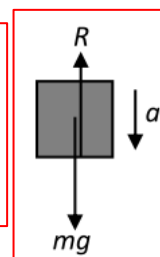
By $F_{\text{net}} = ma$,

$$mg - R = ma$$

$$50 \times 9.81 - R = 50 \times 1$$

$$R = 441 \text{ N}$$

By Newton's third law of motion, the magnitude of the force acting on the weighing scale by Diana is also 441 N.



B

5. Jeanie steps onto a weighing scale in a lift and the reading of the scale is 550 N.

If her mass is 45 kg, which of the following may be happening?

- A The lift is moving up with acceleration 2.41 m s^{-2} .
B The lift is moving down with acceleration 1.8 m s^{-2} .
C The lift is moving at a constant speed of 2.41 m s^{-1} .
D The lift is moving up with deceleration 1.8 m s^{-2} .



A

$$\text{Weight} = 45 \times 9.81 = 441.5 \text{ N}$$

$$F_{\text{net}} = 550 - 441.5 = 108.5 \text{ N (upward)}$$

$$= 45(a)$$

$$a = 2.41 \text{ ms}^{-2}$$

6. Mary, of mass 80 kg, is standing on a weighing scale in a lift. At a certain instant, the reading of the weighing scale is 600 N. Which of the following statements about Mary at this instant is/are correct?

- (1) The gravitational force acting on Mary is 600 N.
(2) The force exerted by the weighing scale on Mary is 600 N.
(3) Mary is accelerating downward at a rate of 2.31 m s^{-2}

A. (2) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

$$mg = R + ma, \text{ the } F_{\text{gravitational}} = (80)(9.81) = 784.8 \text{ N. } R = \text{reading of weighing scale} = 600 \text{ N.}$$

$$a = (mg - R)/m = (784.8 - 600)/80 = 2.31 \text{ ms}^{-2}. \text{ Thus, only (2) and (3) are correct.}$$

D

7. Katie stands on a weighing scale in a lift as shown. Her mass is 50 kg and the mass of the scale is 15 kg. If the reading of the scale is 580 N, what is the magnitude of the force acting on the scale by the lift?

- A 174 N
B 607 N
C 727 N
D 754 N

$$\text{Weight} = 50 \times 9.81 = 490.5 \text{ N}$$

Net force acting on Katie

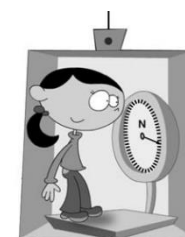
$$= 580 - 490.5 = 89.5 \text{ N}$$

$$a = 89.5/50 = 1.79 \text{ ms}^{-2}$$

Since acceleration of the system is equal to that of Katie

$$R - (50 + 15)(9.81) = (50 + 15)(1.79)$$

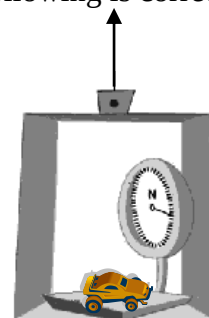
$$R = 754 \text{ N}$$



D

8. A toy car of weight W is placed on a weighing scale that is fixed in a box. The box is pulled up by a rope at a constant speed from time t_1 to t_2 . At t_2 the rope breaks suddenly. Which of the following is correct?

- A Reading = W before t_2 , and then drops to zero gradually.
B Reading $> W$ before t_2 , and then drops to zero gradually.
C Reading = W before t_2 , and then drops to zero suddenly.
D Reading $> W$ before t_2 , and then drops to zero suddenly.



C

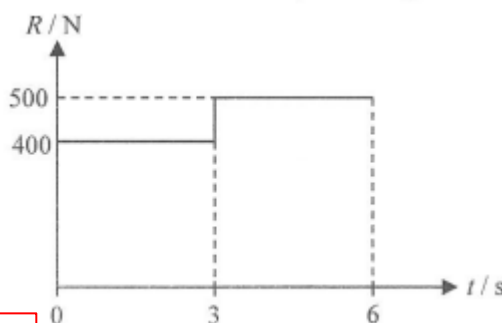
9. A man of mass 50 kg stands still on a balance in a lift which is initially moving downwards. Between $t = 0$ and 6 s, the reading R of the balance varies with time as shown in the graph below. Take g to be 10 m s^{-2} .

B

Which of the following statements is/are correct?

- (1) Between $t = 0$ and 3 s, the net force acting on the man is 400 N.
(2) At $t = 3$ s, the lift starts to move upwards.
(3) Between $t = 3$ s and 6 s, the lift is moving at a constant velocity.

- A. (1) only
B. (3) only
C. (1) & (2) only
D. (2) & (3) only



- * (1) At $t = 0$ to 3 s, $R = 400 \text{ N}$, and $W = mg = 500 \text{ N}$. $\therefore$ Net force = $W - R = 100 \text{ N}$.
* (2) At $t = 3$ s, $R = 500 \text{ N}$, thus there is no net force, and the lift should move down with constant velocity.
✓ (3) At $t = 3$ to 6 s, since there is no net force, thus no acceleration, the lift is moving at constant velocity.

10. A man of mass 50 kg stands still on a weighing scale in a lift. The lift starts to ascend at $t = 0$ and comes to rest at $t = 12$ s. The graph shows how the reading of the weighing scale varies within this time interval. What is the corresponding height ascended by the lift? (Take the acceleration due to gravity to be 10 m s^{-2} .)

C

- A. 48 m
B. 36 m
C. 32 m
D. 24 m

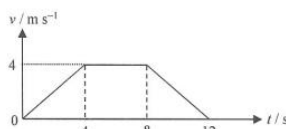
Reading of the scale is equal to the normal reaction R acting on the man.

At $t = 0$ to 4 s, $R = 550 \text{ N}$. By $R - mg = ma$, $(550) - (500) = (50)a$, $a = 1 \text{ m s}^{-2}$. The lift is accelerating upwards.

At $t = 4$ to 8 s, $R = 500 \text{ N}$. As $R = W$, the net force is zero, thus the lift is moving upwards with uniform velocity.

At $t = 8$ to 12 s, $R = 450 \text{ N}$. By $mg - R = ma$, $(500) - (450) = (50)a$, $a = 1 \text{ m s}^{-2}$. The lift is decelerating upwards.

Draw a velocity - time graph as shown.

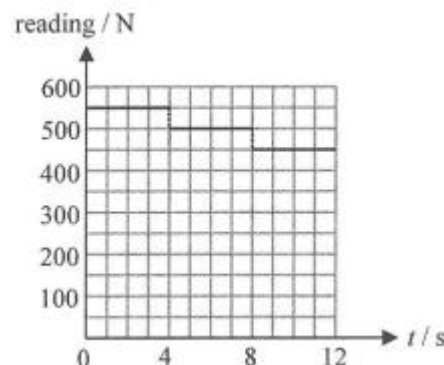


Maximum velocity: $v = u + at = (0) + (1)(4) = 4 \text{ m s}^{-1}$

Or make use of the slope equal to the acceleration.

Height = total distance travelled = area under the graph

$$= \frac{1}{2}(4 + 12)(4) = 32 \text{ m}$$



11. A 50 kg girl stands on a balance in the lift. The following graph shows how her speed varies with time as the lift goes up from G/F to 33/F. The maximum speed of the lift is 3.4 ms^{-1} .

- (a) Draw the labelled force diagram for the girl in the lift.
(b) Calculate the maximum upward acceleration of the lift.
(c) What is the balance reading in the first 2 s?
(d) Calculate the time t_1 for the lift to reach 33/F (100 m above the ground) from the G/F.

(b). $a = \frac{3.4}{2} = 1.7 \text{ ms}^{-2}$

(c). $R - 50(9.81) = 1.7(50)$
 $R = 575.5 \text{ N}$

(d).

$$S_{28-t_1} = 100 - \left(\frac{(28+26)(3.4)}{2} \right)$$

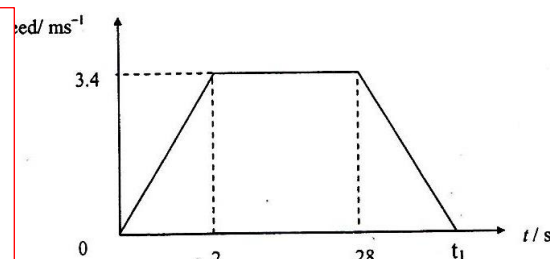
$$= 8.2 \text{ m}$$

$$8.2 = 3.4t - \frac{1}{2} \left(\frac{3.4}{t} \right) t^2$$

$$t = 4.82 \text{ s}$$

$$t_1 = 32.8 \text{ s}$$

警告



12. A girl of mass 60 kg stands on a weighing scale (calibrated in N) in a lift of mass 500 kg (Fig a). Figure b shows the velocity–time graph of the lift. The upward direction is taken to be positive.

The mass of the weighing scale can be neglected.

(a) Find the acceleration of the lift from

- (i) 0–5 s, (ii) 5–20 s, (iii) 20–25 s.

Hence sketch the acceleration–time graph of the lift. (4 marks)

(b) Find the tension in the cable hanging the lift at $t = 3$ s. (2 marks)

(c) What is the reading of the weighing scale from

- (i) 0–5 s,
(ii) 5–20 s,
(iii) 20–25 s.

(4 marks)

(d) If the cable of the lift breaks, the lift undergoes free falling. The reading of the weighing scale becomes zero. Explain briefly why this is the case. (2 marks)

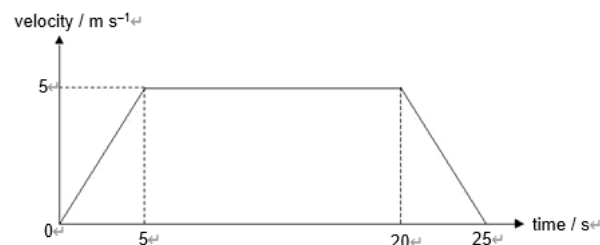


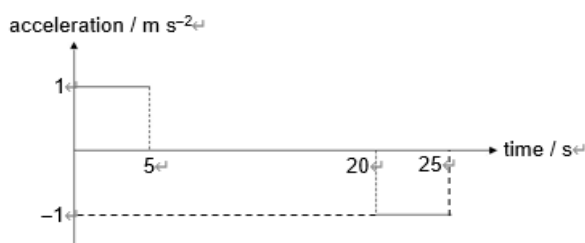
Fig b

(a) Apply acceleration = slope of graph.

(i) Acceleration = $\frac{5-0}{5} = 1 \text{ m s}^{-2}$

(ii) 0

(iii) Acceleration = $\frac{0-5}{25-20} = -1 \text{ m s}^{-2}$



(Correct labelled axes with units)

(Correct graph)

(b) By $F = ma$,

$$T - (60 + 500)9.81 = (60 + 500) \times 1$$

$$T = 6050 \text{ N}$$

The tension is 6050 N.

(c) (i) By $F = ma$,

$$R - 60 \times 9.81 = 60 \times 1$$

$$R = 649 \text{ N}$$

The weighing scale reading is 649 N.

(ii) Weighing scale reading = $mg = 60 \times 9.81 = 589 \text{ N}$

(iii) $R - 60 \times 9.81 = 60 \times (-1)$

$$R = 529 \text{ N}$$

The weighing scale reading is 529 N.

(d) The weighing scale measures the reaction force acting on the girl.

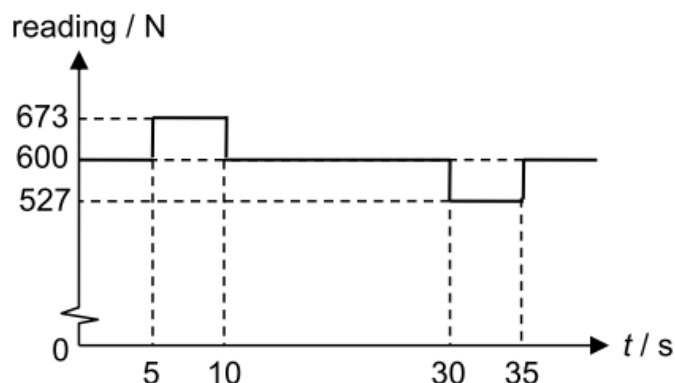
When the girl and the scale undergo free falling, the reaction force acting on her is zero.

Therefore, the scale reading is zero.

必究

13. Sam stands on a weighing scale calibrated in newtons in a lift. The graph below shows how the reading of the scale varies with time t . The lift starts at $t = 5$ s.

- (a) Does the lift go up or down? (1 mark)
(b) Find Sam's acceleration at $t = 8$ s. (2 marks)
(c) Find Sam's acceleration at $t = 32$ s. (2 marks)
(d) Find Sam's velocity at $t = 20$ s. (2 marks)
(e) Sketch Sam's v - t graph from $t = 0$ to $t = 35$ s. (4 marks)
(f) Estimate the distance that Sam travels from $t = 0$ to $t = 35$ s. (2 marks)



Take upwards as positive.

- (a) Go up
(b) Let R be the normal force acting on Sam by the lift.

By $F_{\text{net}} = ma$,

$$R - W = ma$$

$$673 - 600 = \frac{600}{9.81} \times a$$

$$a = 1.194 \text{ m s}^{-2} \approx 1.19 \text{ m s}^{-2}$$

Sam's acceleration is 1.19 m s^{-2} upwards.

- (c) By $F_{\text{net}} = ma$,

$$R - W = ma$$

$$527 - 600 = \frac{600}{9.81} \times a$$

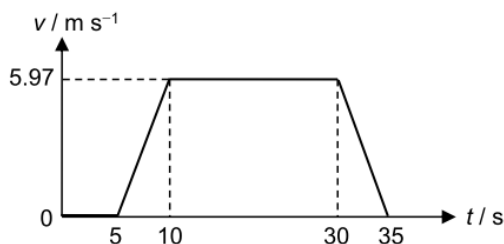
$$a = -1.194 \text{ m s}^{-2} \approx -1.19 \text{ m s}^{-2}$$

Sam's acceleration is 1.19 m s^{-2} downwards.

- (d) $v = u + at = 0 + 1.194 \times (10 - 5)$
 $= 5.97 \text{ m s}^{-1}$

Sam's velocity is 5.97 m s^{-1} upwards.

- (e)



(At rest from 0–5 s)

(Speeding up uniformly from rest from 5–10 s)

(Constant velocity from 10–30 s)

(Slowing down uniformly to rest from 30–35 s)

- (f) Distance travelled = area under v - t graph = $\frac{(30 - 10) + (35 - 5)}{2} \times 5.97$
 $= 149 \text{ m}$

1A

1M

1A

1M

1A

1M

1A

1A

1A

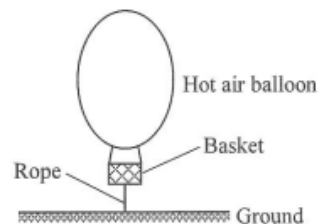
1A

1A

1M

1A

14. A hot-air balloon is tied to the ground by a rope to stop it from rising as shown in the above figure. The total mass of the balloon and the basket is 750 kg. There also exists an upward force of 8000 N acting on the balloon.



- (a) What is the tension of the rope?
(b) The rope is then cut and the balloon start to rise up
(i) What is the acceleration of the balloon after the rope is cut?
(ii) What is the height that the balloon has risen after 1 minute?
(iii) If the balloonist throws some sandbags out of the basket during the rise, what is the effect on the acceleration of the balloon? Explain briefly.

(a) Weight of the balloon = $750 \times 9.81 = 7360 \text{ N}$

$$\therefore (8000) = (7360) + T$$

$$\therefore T = 640 \text{ N} \quad < \text{accept } 643 \text{ N} >$$

(b) (i) Upward force – weight = ma

$$\therefore (8000) - (7360) = (750) a$$

$$\therefore a = 0.853 \text{ m s}^{-2} \quad < \text{accept } 0.840 - 0.865 >$$

(ii) $s = ut + \frac{1}{2} at^2$

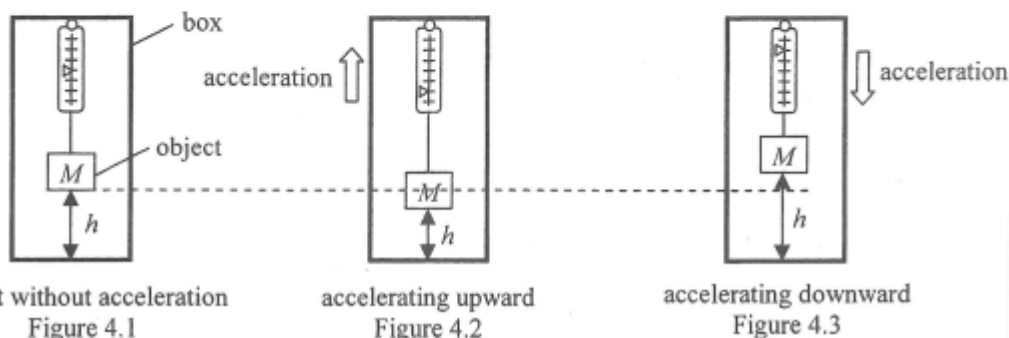
$$= (0) + \frac{1}{2} (0.853) (60)^2 = 1540 \text{ m} \quad < \text{accept } 1500 - 1580 >$$

- (iii) Since the weight of the balloon is reduced,
the upward net force is increased,
thus the upward acceleration is increased.

15.

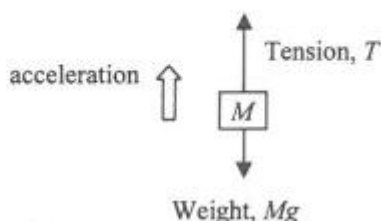
Read the following description about accelerometers and answer the questions that follow.

An accelerometer is a device for measuring acceleration. The following example illustrates the principle of a simple accelerometer. An object of mass M is suspended by a spring balance inside a box. If the box is at rest without acceleration, the object is h above the bottom of the box (Figure 4.1). When the box accelerates upward, h decreases (Figure 4.2). Likewise, when the box accelerates downward, h increases (Figure 4.3). Since it is known that the tension of the spring balance is directly proportional to its extension, we can therefore determine the magnitude and direction of the box's acceleration by measuring h .



- (a) Draw a labelled free-body diagram in the space provided below showing the forces acting on the object when the box accelerates upward with acceleration a . Explain why h decreases in this case. (4 marks)

(a)	Arrows correct	1A
	Labels (tension / T , weight / Mg)	1A
	Or Either force correct and with label	1A
	All correct.	1A



As M accelerates upward,
 $T - Mg = Ma$
i.e. $T = Mg + Ma$

Greater tension needed ($T > \text{original tension} = Mg$) and thus the spring extends further.
i.e. M is closer to the bottom, smaller h in Figure 4.2

1M

1A

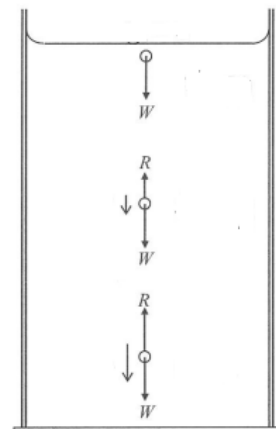
- (b) The scale of the spring balance is calibrated such that the pointer moves 1 cm for a change of 2 N of force. The weight of the object is 5 N. If h decreases by 0.5 cm compared to the situation in Figure 4.1, what is the reading of the balance? Hence find the magnitude of the corresponding acceleration of the box. (acceleration due to gravity $g = 9.81 \text{ m s}^{-2}$) (3 marks)

(b)	Change in tension / reading = $2 \text{ N cm}^{-1} \times 0.5 \text{ cm} = 1 \text{ N}$ $\therefore T - Mg = 1 \text{ N}$ i.e. $T = 6 \text{ N}$	1A
	$T - Mg = Ma$ $1 \text{ N} = \frac{5 \text{ N}}{g} a$ (accept $M = 0.5 \text{ kg}$)	1M
	$a = \frac{1}{5} g$ or $0.2 g$ (upward) (or $\frac{g}{5} = 2 \text{ m s}^{-2}$ or 1.96 m s^{-2})	1A

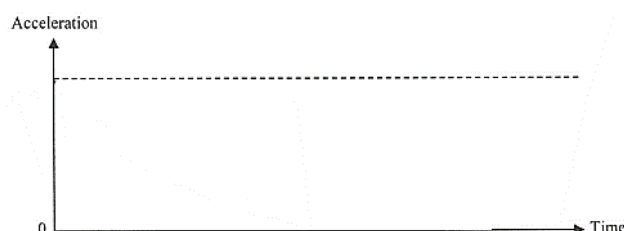
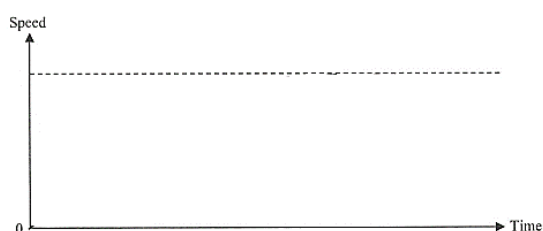
Terminal velocity

Suppose a small ball is dropped from rest in a cylinder containing oil.

- Initially the resistance force R is smaller than the weight W of the ball, the net force on the ball is downwards and the ball accelerates.
- As the velocity v of the ball increases, the resistance force R increases.
- Finally, as the resistance force R just equals the weight W of the ball, the net force on the ball is zero. The ball then moves with uniform velocity, called the **terminal velocity**.

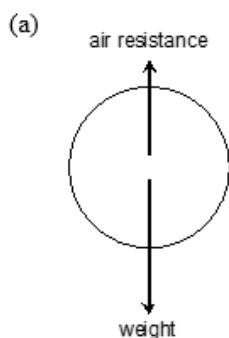
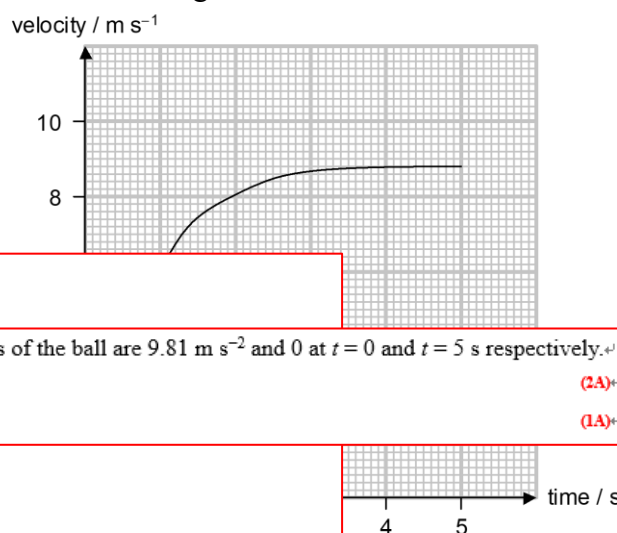


Variation of the speed and acceleration of the ball with time:

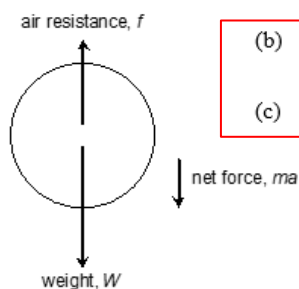


16. The velocity–time graph of a ping-pong ball falling in air is as shown. The ball has a mass of 2 g and is released at $t = 0$.

- Draw a free body diagram to show all the forces acting on the falling ping-pong ball.
- What is the acceleration of the ball at $t = 0$? What is its acceleration at $t = 5$ s?
- What is the terminal velocity of the ball?
- At a certain instant, the acceleration of the ball is 5 m s^{-2} , find the magnitudes of all the force acting on the ball.



(d) The free body diagram is shown below.



The net force acting on the ball is ↓

$$\begin{aligned} F &= ma \\ &= 0.002 \times 5 \\ &= 0.01 \text{ N} \end{aligned}$$

The weight of the ball is ↓

$$W = 0.002 \times 9.81 = 0.01962 \text{ N}$$

Applying Newton's second law of motion, the air resistance acting on the ball is ↓

$$f = F - W = 0.01 - 0.01962 = -0.00962 \text{ N}$$

(b) The accelerations of the ball are 9.81 m s^{-2} and 0 at $t = 0$ and $t = 5$ s respectively.

(c) 8.8 m s^{-1}

Motion of a parachutist falling on a plane

(1) At the beginning:

The skydiver jumping out of an aircraft has zero downward velocity so that he accelerates with $a = g$ as no air resistance acting on him initially and net force acting on him is his weight W .

$$\begin{aligned} \text{By } F_{\text{net}} &= ma \\ \therefore mg &= ma \\ \therefore a &= g \end{aligned}$$

- (2) As he accelerates, his velocity increase and air resistance acting on him increases. The net force acting on him decreases and hence his acceleration decreases.

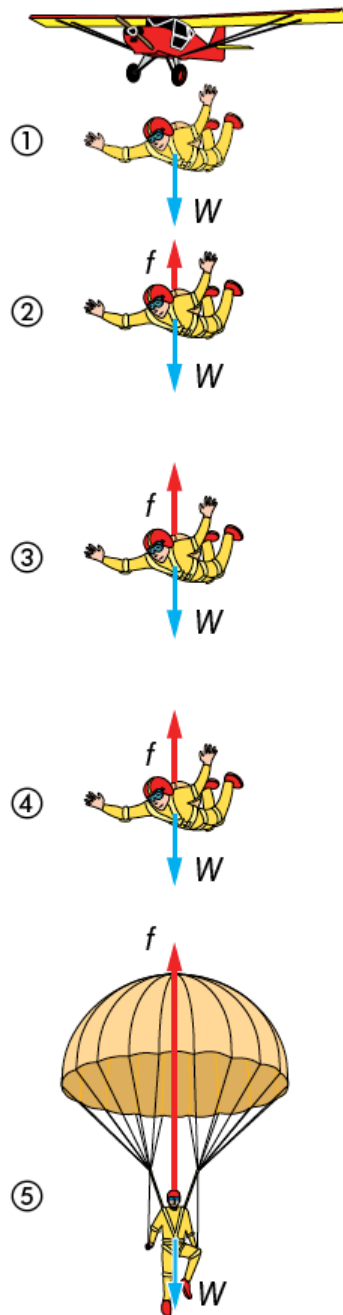
$$\begin{aligned} \text{By } F_{\text{net}} &= ma \\ \therefore mg &> f \\ \therefore mg - f &= ma \end{aligned}$$

- (3) As air resistance f is balanced by W , net force acting on him is equal to zero. The skydiver falls with a **constant speed** called **terminal speed**.

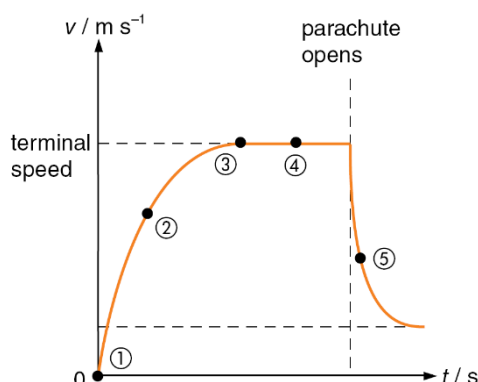
$$\begin{aligned} \text{By } F_{\text{net}} &= ma \\ \therefore mg &= f \\ \therefore a &= 0 \end{aligned}$$

- (4) The skydiver keeps on falling at the terminal speed.

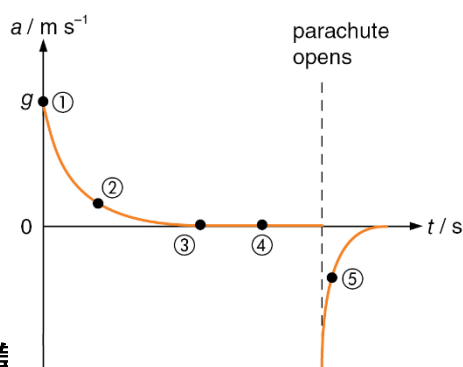
- (5) In order to land safely, the skydiver opens his parachute to increase the air resistance. The net force acting on him ($f - W$) points upwards and hence he decelerates downwards. The speed of the skydiver will be greatly reduced for safety landing.



$v - t$ graph of skydiver



$a - t$ graph of skydiver



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17. A skydiver jumps from an aeroplane and falls for a while before opening her parachute. The graph below shows how the vertical velocity of the skydiver changes after jumping from the aeroplane (taking downwards as positive).

- State the physical meaning of the slope of the graph. (1 mark)
- Describe and explain the motion of the skydiver represented in the velocity–time graph by *AB* and *BC*. (5 marks)
- Describe the motion of the skydiver represented in the velocity–time graph by *CD* and *DE*. (2 marks)
- What does the skydiver do at *C*? (1 mark)
- Find the acceleration of the skydiver from 50 s to 60 s. (2 marks)
- Find the distance travelled by the skydiver from 70 s to 80 s. (2 marks)

(a) Acceleration.

(b) In *AB*:

The skydiver first accelerates under gravity.

The air resistance in the upward direction increases with the velocity of the skydiver.

Therefore, the net downward force and hence acceleration decreases.

In *BC*:

The air resistance just balances the weight of the skydiver. The net force on her is zero.

By Newton's second law of motion, the skydiver falls with a constant velocity (terminal velocity).

(c) In *CD*:

The skydiver decelerates.

In *DE*:

The skydiver falls with a constant velocity.

(d) The skydiver opens her parachute.

(e) Acceleration = $\frac{v - u}{t}$

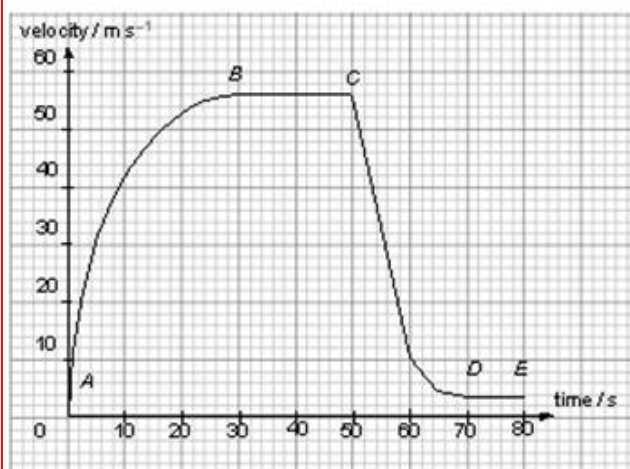
$$= \frac{10 - 56}{60 - 50}$$

$$= -4.6 \text{ m s}^{-2}$$

(f) Distance travelled = vt

$$= 4 \times (80 - 70)$$

$$= 40 \text{ m}$$



18. An object of mass 1 kg is hung by a spring balance inside a box. The following figure shows the whole set-up that is placed on the ground. It is known that the mass of the set-up is 5 kg.

The box is then dropped from a height.

- (a) What is the weight of the object? (2 marks)
- (b) If air resistance is ignored, what is the acceleration of the box as it falls? (1 mark)
- (c) If air resistance should be taken into consideration, how will it affect the acceleration of the box? Explain briefly. (2 marks)
- (d) It is found that the box finally falls with a constant speed.
- (i) Explain why this happens. (3 marks)
- (ii) What is the air resistance when this happens? (1 mark)
- (e) What is the reading of the spring balance when
- (i) the box accelerates at g ?
- (ii) the box attains a constant speed?



Explain your answers briefly.

(5 marks)

(a) Weight of the object = mg

$= 1 \times 9.81$

$= 9.81 \text{ N}$

(b) 9.81 m s^{-2}

- (c) If air resistance is considered, the acceleration will become smaller because air resistance opposes the motion of the box.

- (d) (i) Air resistance increases with the speed of the box. Eventually, it becomes large enough to balance the weight of the box, so the net force acting on the box becomes zero. According to Newton's second law, the box will fall with a constant speed.

(ii) Air resistance = weight of the set-up = $5 \times 9.81 = 49.1 \text{ N}$

- (e) (i) Since the object accelerates at g , the net force acting on it is its weight.

Therefore, the spring balance does not apply any force on it, so its reading is zero.

- (ii) Since the object does not accelerate, the net force acting on it is zero.

Force on object by spring balance = weight of object = 9.81 N

The reading of the spring balance is 9.81 N